DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 8

Electromagnetic waves in vacuum and matter Date: 21 Apr 2025

1. Write the real component of electric and magnetic fields for a monochromatic plane wave (Amplitude $= E_0$, frequency $= \omega$ and phase angle $\delta = 0$) which is
   (a) travelling in the negative x-direction and polarized in the z -direction.
   (b) travelling along ($1, 1, 1$) with polarization parallel to the xz-plane.
2. Consider a linearly polarized plane EM waves propagating in z-directions, with their plane of polarization along the x direction. The electric field's amplitude is given by $|E_0|$, the frequency of the wave is ω, and its wave number is k. Find the value of
$$\frac{\partial U}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{S},$$
where U is the energy density and $\boldsymbol{S}$ is the Poynting vector.
3. The intensity of sunlight hitting the earth is about 1300 W/m^2. If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?
4. A He-Ne laser emits a plane wave which is polarized along $\hat{\boldsymbol{x}}$ and propagating in yz-plane at an angle $\pi/3$ to the y-axis in a medium of refractive index 1.5. The wavelength and intensity of the plane wave are 633 nm and 1 W/m^2, respectively. Calculate the electric and magnetic field associated with the plane waves.
5. Considering an EM wave, traveling in the air, with amplitude 5 V/m and polarized along $\hat{\boldsymbol{y}}$, incident normally on a dielectric of refractive index 2.5. The free space wavelength is 6×10^{-7} m
   (a) Find the reflecting and transmitting waves (i.e., express E_R, H_R, E_T, H_T).
   (b) Calculate the Poynting vectors associated with the incident, reflected, and transmitted wave and show that $R + T = 1$.
6. The refractive index of diamond is 2.42 . Plot the graph of $\frac{E_{0T}}{E_{0I}}$ vs θ_I and $\frac{E_{0R}}{E_{0I}}$ vs θ_I for the air/diamond interface. Here, θ_I is the angle of incidence and consider $\mu_1 = \mu_2 = \mu_0$. Also, calculate
   (a) the amplitude of normal incidence,
   (b) Brewster's Angle, and
   (c) the angle at which the reflected and transmitted amplitudes are equal.
7. **The skin effect:** Recall the relations $\mathbf{D} = \epsilon\mathbf{E}, \mathbf{B} = \mu\mathbf{H}, \mathbf{J}_f = \sigma\mathbf{E}$ (Ohm's Law). For a metal under normal circumstances, $\mathbf{J}_f$ is much larger than $\partial\mathbf{D}/\partial t$.
   (a) Neglect ρ_f and show that for a metal, $\mathbf{E}$ satisfies the equation
$$\nabla^2\mathbf{E} = \mu\sigma\frac{\partial\mathbf{E}}{\partial t}$$

(b) Consider a "*plane wave*" solution of the above equation of the form $\mathbf{E} = \mathbf{E}_0 \exp i(kz - \omega t)$, for $z > 0$. Find the allowed values of the wave number k as a function of the frequency ω.

(c) Interpret the form of the solution. How does the amplitude of the electric field vary with k, and at what distance does it decay to $1/e$ of its value at $z = 0$?

Extra Questions:

8. An infinitely long straight non-magnetic conductor (wire) with a circular cross-section of radius a carries a steady current I. The current is distributed uniformly over the cross-section of the wire. The conductivity of the wire is σ. Find the rate at which energy flows into unit length of the conductor.

9. A beam of protons has a circular cross-section. Each proton has a velocity $\mathbf{v}$, and the beam constitutes a current I. Find the direction and magnitude of the Poynting vector $\mathbf{S}$ outside the beam, at a distance r from the axis of the beam.

1. Write the real component of electric and magnetic fields for a monochromatic plane wave (Amplitude $= E_0$, frequency $= \omega$ and phase angle $\delta = 0$) which is
   (a) travelling in the negative x-direction and polarized in the z -direction.
   (b) travelling along ($1, 1, 1$) with polarization parallel to the xz-plane.

(a) $\hat{k} = -\hat{\imath}$; $\hat{n} = \hat{k}$

$$\bar{E}(\bar{r}, t) = E_0 \hat{k} \cos\left((\hat{\imath}).(\bar{r}) + \omega t\right)$$

$$\bar{B}(\bar{r}, t) = \frac{E_0}{c} \hat{\jmath} \cos\left((\hat{\imath}).(\bar{r}) + \omega t\right)$$

(b) $\hat{k} = \frac{\hat{\imath}+\hat{\jmath}+\hat{k}}{\sqrt{3}}$; $\hat{n} = -\frac{\hat{\imath}-\hat{k}}{\sqrt{2}}$

$$\bar{E}(\bar{r}, t) = E_0 \left(\frac{-\hat{\imath}-\hat{k}}{\sqrt{2}}\right) \cos\left(\left(\frac{\hat{\imath}+\hat{\jmath}+\hat{k}}{\sqrt{3}}\right).\bar{r} - \omega t\right)$$

$$\bar{B}(\bar{r}, t) = \frac{E_0}{c} \left(\frac{-\hat{\imath}+\hat{k}}{\sqrt{2}}\right) \cos\left(\left(\frac{\hat{\imath}+\hat{\jmath}+\hat{k}}{\sqrt{3}}\right).\bar{r} - \omega t\right)$$

2. Consider a linearly polarized plane EM waves propagating in z-directions, with their plane of polarization along the x direction. The electric field's amplitude is given by $|E_0|$, the frequency of the wave is ω, and its wave number is k. Find the value of

$$\frac{\partial U}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{S},$$

where U is the energy density and $\boldsymbol{S}$ is the Poynting vector.

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0} = \varepsilon_0 E^2 = \varepsilon_0 |E_0|^2 \cos^2(\bar{k}.\bar{r} - \omega t + \delta)$$

$$\bar{S} = \frac{1}{\mu_0}(\bar{E} \times \bar{B}) = \frac{1}{\mu_0} \frac{|E_0|^2}{c} \cos^2(\bar{k}.\bar{r} - \omega t + \delta)\,\hat{z} = c.u.\hat{z}$$

$$\bar{E}(\bar{r},t) = |E_0|\hat{x}\cos(\bar{k}\cdot\bar{r} - \omega t + \delta)$$

$$\bar{B}(\bar{r},t) = \frac{|E_0|\hat{y}}{c}\cos(\bar{k}\cdot\bar{r} - \omega t + \delta)$$

$$\frac{du}{dt} = \varepsilon_0 |E_0|^2 2\cos(\bar{k}\cdot\bar{r} - \omega t + \delta)\cdot(-\sin(\bar{k}\cdot\bar{r} - \omega t + \delta))(-\omega)$$

$$\bar{\nabla}\cdot\bar{S} = c\cdot\frac{du}{dz} = c\cdot\varepsilon_0|E_0|^2\cdot 2\cos(\bar{k}\cdot\bar{r} - \omega t + \delta)(-\sin(\bar{k}\cdot\bar{r} - \omega t + \delta))(k) = -\frac{du}{dt}$$

$$\Rightarrow \bar{\nabla}\cdot\bar{S} + \frac{du}{dt} = 0$$

3. The intensity of sunlight hitting the earth is about 1300 W/m^2. If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?

$$I = 1300\ W/m^2 = \langle S\rangle = \frac{1}{2}c\varepsilon_0 E_0^2$$

$$g\ (\text{momentum energy density}) = \frac{S}{c^2}$$

$$\Rightarrow \langle g\rangle = \frac{\langle S\rangle}{c^2} = \frac{1}{2c}\varepsilon_0 E_0^2$$

$$\Delta P = \langle g\rangle \times A c \times \Delta T = \frac{A\varepsilon_0 E_0^2 \Delta T}{2}$$

$$P = \frac{F}{A} = \frac{\Delta P}{A\Delta T} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{I}{c}$$

absorber: $\frac{2600\ W/m^2}{3\times10^8\ m/s} = 8.67\times10^{-6}\ Pa$

Fraction with Patm $= 8.67\times10^{-11}$

reflector: momentum change doubles

$\Rightarrow P = 17.33 \times 10^{-6}$ Pa

fraction $= 17.33 \times 10^{-11}$ with Patm

4. A He-Ne laser emits a plane wave which is polarized along $\hat{\boldsymbol{x}}$ and propagating in yz-plane at an angle $\pi/3$ to the y-axis in a medium of refractive index 1.5. The wavelength and intensity of the plane wave are 633 nm and 1 W/m^2, respectively. Calculate the electric and magnetic field associated with the plane waves.

$\hat{n} = \uparrow \qquad \hat{k} = \frac{\hat{j}}{2} + \frac{\hat{k}\sqrt{3}}{2} \qquad \mu = 1.5 = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$

$\lambda = 633 \times 10^{-9}$ m $\qquad \approx \sqrt{\varepsilon_r}$

$I = 1 W/m^2 \qquad \Rightarrow \varepsilon_r = 2.25$

$1 W/m^2 = \frac{1}{2} \times \frac{1}{36\pi \times 10^{-9}} \times 2.25 \times |E_0|^2 \times 3 \times 10^8$

$|E_0| = 1.353 \times 10^{-4}$

$\bar{E}_0 = 1.353 \times 10^{-4} \cos(\bar{k}\cdot\bar{r} - \omega t)\uparrow$

$\bar{B}_0 = 4.51 \times 10^{-13} \cos(\bar{k}\cdot\bar{r} - \omega t)\left(\frac{\sqrt{3}}{2}\uparrow - \frac{1}{2}\hat{j}\right)$

5. Considering an EM wave, traveling in the air, with amplitude 5 V/m and polarized along $\hat{\boldsymbol{y}}$, incident normally on a dielectric of refractive index 2.5. The free space wavelength is 6×10^{-7} m
   (a) Find the reflecting and transmitting waves (i.e., express E_R, H_R, E_T, H_T).
   (b) Calculate the Poynting vectors associated with the incident, reflected, and transmitted wave and show that $R + T = 1$.

prop $\rightarrow \hat{z}$
$E \rightarrow \hat{y}$
$B \rightarrow (-\hat{x})$

(a) $E_R = E_{0R}\, e^{i(-k_1 z - \omega t)}\hat{y}$

$B_R = \frac{E_{0R}}{c}\, e^{i(-k_1 z - \omega t)}\hat{x}$

$$E_T = E_{0T}\, e^{i(k_2 z - \omega t)}\, \hat{y}$$

$$B_T = \frac{E_{0T}}{c}\, e^{i(k_2 z - \omega t)}\, (-\hat{x})$$

$$\left[\bar{H} = \frac{\bar{k} \times \bar{E}}{\omega \mu_0}\right]$$

$$E_{0R} = \frac{3}{7} E_0 \qquad E_{0T} = \frac{4}{7} E_0$$

(b) $I = \frac{1}{\mu_0}(\bar{E} \times \bar{B})$

$$R = \frac{1}{\mu_0}(\bar{E}_R \times \bar{B}_R) = \frac{9}{49}$$

$$T = \frac{1}{\mu_0}(\bar{E}_T \times \bar{B}_T) = \frac{16}{49} \times 2.5$$

$$R + T = 1$$

7. **The skin effect:** Recall the relations $\mathbf{D} = \epsilon\mathbf{E}, \mathbf{B} = \mu\mathbf{H}, \mathbf{J}_f = \sigma\mathbf{E}$ (Ohm's Law). For a metal under normal circumstances, $\mathbf{J}_f$ is much larger than $\partial\mathbf{D}/\partial t$.

(a) Neglect ρ_f and show that for a metal, $\mathbf{E}$ satisfies the equation

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t}$$

$$\bar{\nabla} \cdot \bar{E} = 0$$

$$\bar{\nabla} \cdot \bar{B} = 0$$

$$\bar{\nabla} \times \bar{E} = -\partial B / \partial t$$

$$\bar{\nabla} \times \bar{B} = \mu_0 J_f + \mu_0 \varepsilon_0 \frac{\partial \bar{E}}{\partial t}$$

$$\bar{\nabla} \times \bar{B} = \mu_0 \sigma E + \mu_0 \varepsilon_0 \frac{\partial \bar{E}}{\partial t}$$

$$\bar{\nabla} \times (\bar{\nabla} \times \bar{E}) = \bar{\nabla}(\bar{\nabla} \cdot \bar{E}) - \nabla^2 E = -\frac{\partial}{\partial t}\left(\mu_0 \sigma E + \mu_0 \varepsilon_0 \frac{\partial \bar{E}}{\partial t}\right)$$

$$\Rightarrow \nabla^2 E = \mu_0 \sigma \frac{\partial E}{\partial t} + \mu_0 \varepsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\Rightarrow \nabla^2 \bar{E} = \mu_0 \sigma \frac{\partial \bar{E}}{\partial t}$$

small for metals

(b) Consider a "*plane wave*" solution of the above equation of the form $\mathbf{E} = \mathbf{E}_0 \exp i(kz - \omega t)$, for $z > 0$. Find the allowed values of the wave number k as a function of the frequency ω.

$E = E_0 e^{i(kz-\omega t)} \Rightarrow -k^2 E = \mu_0 \sigma (-i\omega) E \Rightarrow k^2 = \mu_0 \sigma i\omega$

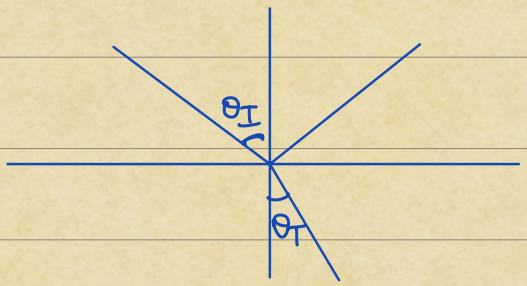
$k = \alpha + i\beta \Rightarrow \alpha^2 - \beta^2 = 0, 2\alpha\beta = \mu_0 \sigma \omega \Rightarrow \alpha = (\mu_0 \sigma \omega / 2)^{1/2}$

(c) Interpret the form of the solution. How does the amplitude of the electric field vary with k, and at what distance does it decay to $1/e$ of its value at $z = 0$?

$E = E_0 e^{i(kz-\omega t)} = E_0 \cos(kz - \omega t)$

6. The refractive index of diamond is 2.42 . Plot the graph of $\frac{E_{0T}}{E_{0I}}$ vs θ_I and $\frac{E_{0R}}{E_{0I}}$ vs θ_I for the air/diamond interface. Here, θ_I is the angle of incidence and consider $\mu_1 = \mu_2 = \mu_0$. Also, calculate
   (a) the amplitude of normal incidence,
   (b) Brewster's Angle, and
   (c) the angle at which the reflected and transmitted amplitudes are equal.

(b) $\tan^{-1}(2.42) = \theta_B = 67.55^\circ$



(a) $R_{11} = \frac{E_{0,R}}{E_{0,I}} = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t}$

$T_{11} = \frac{E_{0,T}}{E_{0,I}} = \frac{2n_i \cos\theta_i}{n_t \cos\theta_i + n_i \cos\theta_t}$

$n_i \sin\theta_I = n_T \sin\theta_T$

(c) equate R_{11} and T_{11}

8. An infinitely long straight non-magnetic conductor (wire) with a circular cross-section of radius a carries a steady current I. The current is distributed uniformly over the cross-section of the wire. The conductivity of the wire is σ. Find the rate at which energy flows into unit length of the conductor.

$$B \times 2\pi r = \mu_0 \frac{I}{\pi a^2} \pi r^2 \Rightarrow \vec{B} = \frac{\mu_0 I r}{2\pi a^2} \hat{\phi}$$

$$J = \sigma \vec{E} \Rightarrow \frac{I}{\pi a^2} = \sigma \vec{E} \Rightarrow \vec{E} = \frac{I}{\pi \sigma a^2} \hat{z}$$

$$\vec{S} = \frac{1}{\mu_0} \left(\frac{\mu_0 I^2 r}{2\pi^2 \sigma a^4} \right) (-\hat{r}) = \frac{I^2 r}{2\pi^2 \sigma a^4} (-\hat{r})$$

$$S_{r=a} = a = \frac{-I^2}{2\pi^2 \sigma a^3} \hat{r}$$

$$P = \int \vec{S} \cdot d\vec{a} = \frac{-I^2}{2\pi^2 \sigma a^3} \cdot 2\pi a l = \frac{-I^2 l}{\pi \sigma a^2} = -I^2 \left(\frac{l}{A} \right) = -I^2 R$$

$$P/l = \frac{-I^2}{\pi \sigma a^2}$$

9. A beam of protons has a circular cross-section. Each proton has a velocity $\mathbf{v}$, and the beam constitutes a current I. Find the direction and magnitude of the Poynting vector $\mathbf{S}$ outside the beam, at a distance r from the axis of the beam.

$$\vec{E} = \frac{\lambda}{2\pi \varepsilon_0 r} \hat{r} \qquad B \cdot 2\pi r = \mu_0 (I)$$

$$\lambda = I/v \qquad \Rightarrow B = \frac{\mu_0 I}{2\pi r} \hat{\phi}$$

$$E = \frac{I}{2\pi \varepsilon_0 r v} \Rightarrow S = \frac{I^2}{4\pi^2 \varepsilon_0 v r^2} \hat{z}$$

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Problem Set 8 - Solutions

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1. Problem-1

Question: Write the real component of electric and magnetic fields for a monochromatic plane wave (Amplitude $= E_0$, frequency $= \omega$ and phase angle $\delta = 0$) which is

(a) travelling in the negative x-direction and polarized in the z -direction.

(b) travelling along ($1, 1, 1$) with polarization parallel to the xz-plane.

Solution:
As given in the question

$$\text{Amplitude } = E_0, \quad \text{Frequency } = \omega \quad \text{and} \quad \text{Phase angle}(\delta) = 0$$

The real part of the electric and the magnetic field, as discussed in Griffith's Book, is

$$\boldsymbol{E_R}(r,t) = \boldsymbol{E_0}\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\hat{\boldsymbol{n}}$$
$$\boldsymbol{B_R}(r,t) = \frac{\boldsymbol{E_0}}{c}\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)(\hat{\boldsymbol{k}}\times\hat{\boldsymbol{n}})$$

(a) As the wave is traveling in the x -direction

$$\boldsymbol{k} = \frac{\omega}{c}(-\hat{\boldsymbol{x}}) \quad \text{and} \quad \hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}.$$

Now, it is easy to verify $\boldsymbol{k}.\boldsymbol{r} = -\frac{\omega}{c}x$ and $\hat{\boldsymbol{k}}\times\hat{\boldsymbol{n}} = \hat{\boldsymbol{y}}$. The real part of the electric and magnetic field is

$$\boldsymbol{E_R}(r,t) = \boldsymbol{E}_0\cos\left(\frac{\omega}{c}[x+ct]\right)\hat{\boldsymbol{z}}$$
$$\boldsymbol{B_R}(r,t) = \frac{\boldsymbol{E_0}}{c}\cos\left(\frac{\omega}{c}[x+ct]\right)\hat{\boldsymbol{y}}$$

(b) As wave travelling along $(1, 1, 1)$ then [1],

$$\boldsymbol{k} = \frac{\omega}{c}\left(\frac{\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}+\hat{\boldsymbol{z}}}{\sqrt{3}}\right) \quad \text{and} \quad \hat{\boldsymbol{n}} = \frac{\hat{\boldsymbol{x}}-\hat{\boldsymbol{z}}}{\sqrt{2}}.$$

Now, it is easy to verify $\boldsymbol{k}.\boldsymbol{r} = \frac{\omega}{\sqrt{3}c}(x+y+z)$ and $\hat{\boldsymbol{k}}\times\hat{\boldsymbol{n}} = \frac{1}{\sqrt{6}}(-\hat{\boldsymbol{x}}+\boldsymbol{2}\hat{\boldsymbol{y}}-\hat{\boldsymbol{z}})$. The real part of the electric and magnetic field is

$$\boldsymbol{E_R}(r,t) = \boldsymbol{E}_0\cos\left(\frac{\omega}{\sqrt{3}c}[x+y+z-\sqrt{3}ct]\right)\left(\frac{\hat{\boldsymbol{x}}-\hat{\boldsymbol{z}}}{\sqrt{2}}\right)$$
$$\boldsymbol{B_R}(r,t) = \frac{\boldsymbol{E_0}}{c}\cos\left(\frac{\omega}{\sqrt{3}c}[x+y+z-\sqrt{3}ct]\right)\left(\frac{-\hat{\boldsymbol{x}}+\boldsymbol{2}\hat{\boldsymbol{y}}-\hat{\boldsymbol{z}}}{\sqrt{6}}\right)$$

[01] While deriving $\hat{\boldsymbol{n}}$ parallel to xz-plane we start with the general expression of $\hat{\boldsymbol{n}} = \alpha\hat{\boldsymbol{x}} + \beta\hat{\boldsymbol{z}}$ and using the condition $\hat{\boldsymbol{n}}.\boldsymbol{k} = 0$ results in $\alpha = -\beta$.

2. Problem-2

Question: Consider a linearly polarized plane EM waves propagating in z-directions, with their plane of polarization along the x direction. The electric field's amplitude is given by $|E_0|$, the frequency of the wave is ω, and its wave number is k. Find the value of

$$\frac{\partial U}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{S},$$

where U is the energy density and $\boldsymbol{S}$ is the Poynting vector.
Solution:

For the EM wave having arbitrary phase ϕ is

$$\boldsymbol{E} = |\boldsymbol{E_0}| \cos(kz - wt + \phi)\hat{\boldsymbol{x}} \quad \text{and} \quad \boldsymbol{B} = \frac{k\,|\boldsymbol{E_0}|}{\omega} \cos(kz - wt + \phi)\hat{\boldsymbol{y}}.$$

As we know, the energy density can be written as

$$U = \frac{\epsilon_0}{2}|\boldsymbol{E}|^2 + \frac{1}{2\mu_0}|\boldsymbol{B}|^2.$$

For the above solution

$$U = \epsilon_0 \left|\boldsymbol{E_0}\right|^2 \cos^2(kz - wt + \phi)$$

and

$$\frac{\partial U}{\partial t} = \omega\epsilon_0 \left|\boldsymbol{E_0}\right|^2 \sin(2[kz - wt + \phi]) \tag{1}$$

Now the Poynting vector $\boldsymbol{S}$ for first wave is

$$\begin{aligned} \boldsymbol{S} &= \frac{\boldsymbol{E} \times \boldsymbol{B}}{\mu_0} \\ &= \frac{k\left|\boldsymbol{E_0}\right|^2}{\mu_0\omega} \cos^2(kz - wt + \phi)\hat{\boldsymbol{z}} \\ \boldsymbol{\nabla}.\boldsymbol{S} &= -\frac{k^2\left|\boldsymbol{E_0}\right|^2}{\mu_0\omega} \sin(2[kz - wt + \phi]). \end{aligned} \tag{2}$$

By adding eq.(1) and eq.(2) we get

$$\frac{\partial U}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{S} = 0$$

3. Problem-3

Question: The intensity of sunlight hitting the earth is about 1300 W/m^2. If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?
Solution:
As we know, the pressure exerted is

$$P = \frac{I}{c} = \frac{1.3 \times 10^3}{3 \times 10^8} = 4.3 \times 10^{-6}\ \text{N/m}^2$$

For a perfect reflector, the pressure is $P = 8.6 \times 10^{-6}$ N/m^2. Fraction of atmospheric pressure amounts in

$$\frac{\text{Pressure}}{\text{Atmospheric Pressure}} = \frac{8.6 \times 10^{-6}}{1.03 \times 10^{5}} \approx 8.3 \times 10^{-11}$$

Note : Please follow the discussion made in section: 9.2.3

4. Problem-4

Question: A He-Ne laser emits a plane wave which is polarized along $\hat{\boldsymbol{x}}$ and propagating in yz-plane at an angle $\pi/3$ to the y-axis in a medium of refractive index 1.5. The wavelength and intensity of the plane wave are 633 nm and 1 W/m^2, respectively. Calculate the electric and magnetic field associated with the plane waves.

Solution:

Given,

$$\lambda = 6.33 \times 10^{-7} \text{ m}$$
$$\text{Intensity } = 1 \text{ W/m}^2$$

It is easy to verify $E_0 \approx 33.62$ V/m.

As the wave is propagating in the yz-plane with an angle $\pi/3$ to the y-axis

$$\begin{aligned} \boldsymbol{k} &= \frac{2\pi}{\lambda}\left[\cos\frac{\pi}{3}\hat{\boldsymbol{y}} + \sin\frac{\pi}{3}\hat{\boldsymbol{z}}\right] \\ &= 7.445 \times 10^{6}[\hat{\boldsymbol{y}} + \sqrt{3}\hat{\boldsymbol{z}}] \end{aligned}$$

The associated E-field can be written as

$$\boldsymbol{E} = \boldsymbol{E_0} e^{i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)}$$

So, finally

$$\begin{aligned} \boldsymbol{E}_{\text{Real}} &= 33.62\left\{\cos\left[7.45 \times 10^{6}(\hat{\boldsymbol{y}} + \sqrt{3}\hat{\boldsymbol{z}})\cdot\boldsymbol{r} - 2.98 \times 10^{5}t\right]\right\}\hat{\boldsymbol{x}} \\ \boldsymbol{H}_{\text{Real}} &= 0.062\left\{\cos\left[7.45 \times 10^{6}(\hat{\boldsymbol{y}} + \sqrt{3}\hat{\boldsymbol{z}})\cdot\boldsymbol{r} - 2.98 \times 10^{5}t\right]\right\}(\sqrt{3}\hat{\boldsymbol{y}} - \hat{\boldsymbol{z}}) \end{aligned}$$

Note: While deriving $\boldsymbol{H}$, we have use the formula $\boldsymbol{H} = \frac{\boldsymbol{K}\times\boldsymbol{E}}{\omega\mu_0}$

5. Problem-5

Question: Considering an EM wave, traveling in the air, with amplitude 5 V/m and polarized along $\hat{\boldsymbol{y}}$, incident normally on a dielectric of refractive index 2.5. The free space wavelength is 6×10^{-7} m

(a) Find the reflecting and transmitting waves (i.e., express E_R, H_R, E_T, H_T).

(b) Calculate the Poynting vectors associated with the incident, reflected, and transmitted wave and show that $R + T = 1$.

Solution:

As we know,

$$\boldsymbol{E_{0R}} = \left|\frac{n_1 - n_2}{n_1 + n_2}\right| \boldsymbol{E_{0I}} = 0.429\boldsymbol{E_{0I}}$$
$$\boldsymbol{E_{0T}} = \frac{2n_1}{n_1 + n_2}\boldsymbol{E_{0I}} = 0.571\boldsymbol{E_{0I}}$$

According to the question

$$\boldsymbol{E}_{0I} = 5 \text{ V/m} \quad ; \quad \hat{\boldsymbol{n}} = \hat{\boldsymbol{y}} \quad ; \quad \lambda = 6 \times 10^{-7} \text{ m}.$$

By using the formulae: $\boldsymbol{B_{0R}} = \frac{\boldsymbol{E_{0R}}}{c}, \boldsymbol{H_{0R}} = \frac{\boldsymbol{B_{0R}}}{\mu_0}$ and $\boldsymbol{H_{0T}} = \frac{\boldsymbol{E_{0T}}}{\boldsymbol{v}}$. Again taking only real components can be written as

- **Incident Wave:**

$$\boldsymbol{E_I} = 5\left\{\cos\left(\boldsymbol{k_1 z} - \omega t\right)\right\}\hat{\boldsymbol{y}} \quad ; \quad \boldsymbol{H_I} = 1.33 \times 10^{-2}\left\{\cos\left(\boldsymbol{k_1 z} - \omega t\right)\right\} \quad \hat{\boldsymbol{z}}$$

- **Transmitted Wave:**

$$\boldsymbol{E_T} = 2.86\left\{\cos\left(\boldsymbol{k_2 z} - \omega t\right)\right\} \quad \hat{\boldsymbol{y}} \quad ; \quad \boldsymbol{H_T} = 1.9 \times 10^{-2}\left\{\cos\left(\boldsymbol{k_2 z} - \omega t\right)\right\} \quad \hat{\boldsymbol{z}}$$

- **Reflected Wave:**

$$\boldsymbol{E_R} = 2.146\left\{\cos\left(\boldsymbol{k_1 z} + \omega t\right)\right\}\hat{\boldsymbol{y}} \quad ; \quad \boldsymbol{H_R} = 5.7 \times 10^{-3}\left\{\cos\left(\boldsymbol{k_2 z} + \omega t\right)\right\} \quad \hat{\boldsymbol{z}}$$

where $k_1 = \frac{\omega}{c}n_1$ and $k_2 = \frac{\omega}{c}n_2$. Now the Poynting vector

$$\langle \boldsymbol{S_I} \rangle = \langle \boldsymbol{E_I} \times \boldsymbol{H_I} \rangle = 3.325 \times 10^{-2} \quad \text{J/m}^2$$
$$\langle \boldsymbol{S_T} \rangle = \langle \boldsymbol{E_T} \times \boldsymbol{H_T} \rangle = 2.717 \times 10^{-2} \text{ J/m}^2$$
$$\langle \boldsymbol{S_R} \rangle = \langle \boldsymbol{E_R} \times \boldsymbol{H_R} \rangle = -6.105 \times 10^{-3} \text{ J/m}^2$$

Now,

$$R = \frac{\langle S_R \rangle}{\langle S_I \rangle} \approx 0.183 \quad ; \quad T = \frac{\langle S_T \rangle}{\langle S_I \rangle} \approx 0.817$$

It is easy to see $R + T = 1$.

6. Problem-6

Question: The refractive index of diamond is 2.42 . Plot the graph of $\frac{E_{0T}}{E_{0I}}$ vs θ_I and $\frac{E_{0R}}{E_{0I}}$ vs θ_I for the air/diamond interface. Here, θ_I is the angle of incidence and consider $\mu_1 = \mu_2 = \mu_0$. Also, calculate

(a) the amplitude of normal incidence,

(b) Brewster's Angle, and

(c) the angle at which the reflected and transmitted amplitudes are equal.

Solution:

Some of the equations already discussed in the class are

$$\alpha = \frac{1}{\cos\theta_I}\sqrt{1-\left[\frac{n_1}{n_2}\sin\theta_I\right]^2} \quad ; \quad \frac{E_{0R}}{E_{0I}} = \left(\frac{\alpha-\beta}{\alpha+\beta}\right) \quad ; \quad \frac{E_{0T}}{E_{0I}} = \left(\frac{2}{\alpha+\beta}\right),$$

where α and β are amplitudes.

(a) As given $n_2 = 2.42$, it is easy to see $\beta = 2.42$.

$$\alpha = \frac{1}{\cos\theta_I}\sqrt{1-\left[\frac{1}{2.42}\sin\theta_I\right]^2}.$$

For $\theta_I = 0$

$$\frac{E_{0R}}{E_{0I}} = \left(\frac{\alpha-\beta}{\alpha+\beta}\right) = -0.415 \quad ; \quad \frac{E_{0T}}{E_{0I}} = \left(\frac{2}{\alpha+\beta}\right) = 0.585$$

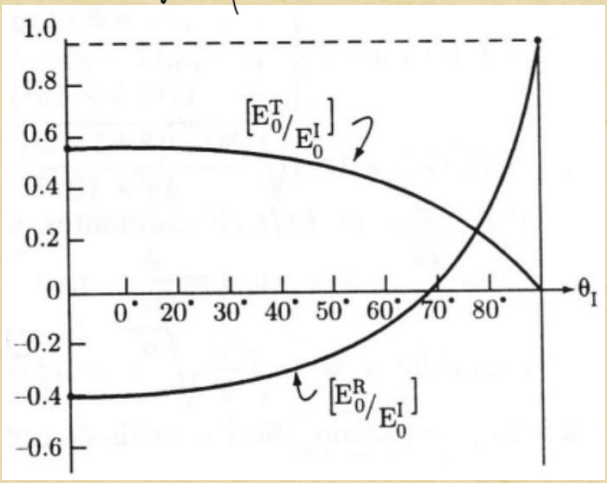


Figure 1: graph of $\frac{E_{0T}}{E_{0I}}$ vs θ_I and $\frac{E_{0R}}{E_{0I}}$ vs θ_I.

(b) Using the relation for Brewster's Angle

$$\theta_B = \tan^{-1}(2.42) \approx 67.5^\circ$$

(c) For $E_{0R} = E_{0T}$ we have

$$\alpha - \beta = 2$$
$$\alpha = 4.42.$$

Now using the above relation

$$4.42 = \frac{1}{\cos\theta_I}\sqrt{1-\left[\frac{1}{2.42}\sin\theta_I\right]^2}$$

By solving for θ_I it is easy to verify $\theta_I \approx 78.3^\circ$

7. Problem-7

Question: The skin effect: Recall the relations $\mathbf{D} = \epsilon\mathbf{E}, \mathbf{B} = \mu\mathbf{H}, \mathbf{J}_f = \sigma\mathbf{E}$ (Ohm's Law). For a metal under normal circumstances, $\mathbf{J}_f$ is much larger than $\partial\mathbf{D}/\partial t$.

(a) Neglect ρ_f and show that for a metal, $\mathbf{E}$ satisfies the equation

$$\nabla^2\mathbf{E} = \mu\sigma\frac{\partial\mathbf{E}}{\partial t}$$

(b) Consider a "*plane wave*" solution of the above equation of the form $\mathbf{E} = \mathbf{E}_0 \exp i(kz - \omega t)$, for $z > 0$. Find the allowed values of the wave number k as a function of the frequency ω.

(c) Interpret the form of the solution. How does the amplitude of the electric field vary with k, and at what distance does it decay to $1/e$ of its value at $z = 0$?

Solution:

(a) We begin with the Maxwell equation curl $\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$ and compute the curl of both sides of the equation.

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{E}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{E}) - \nabla^2\mathbf{E} \simeq -\nabla^2\mathbf{E}$$
$$-\boldsymbol{\nabla} \times \frac{\partial\mathbf{B}}{\partial t} = -\mu\frac{\partial}{\partial t}\left(4\pi J_f + \frac{\partial\mathbf{D}}{\partial t}\right) \simeq -\mu\sigma\frac{\partial\mathbf{E}}{\partial t}$$

where we have implemented the approximations that were suggested in the statement of the problem. Equating the expressions at the end of the two lines above, we get

$$\nabla^2\mathbf{E} = \mu\sigma\frac{\partial\mathbf{E}}{\partial t}$$

(b) Substituting for $\mathbf{E} = \mathbf{E}_0 \exp i(kz - \omega t)$ in the above equation [1], we get

$$-k^2\mathbf{E}_0 = (-i\omega\mu\sigma)\mathbf{E}_0$$

Clearly for the above equation to make sense, we need to solve the above equation for complex k. We get

$$k = \pm\frac{1+i}{\sqrt{2}}\sqrt{\omega\mu\sigma}$$

We think of the region $z > 0$ to be a conducting region with a plane wave being incident at $z = 0$. The electric field takes the form, for $z > 0$, for which we choose the positive sign above

$$\mathbf{E} = \mathbf{E}_0 \exp\left[i\left(\sqrt{\frac{\mu\sigma\omega}{2}}z - \omega t\right)\right] \exp\left[-\sqrt{\frac{\mu\sigma\omega}{2}}z\right]$$

Taking the real part of the above equation, we obtain

$$\mathbf{E} = \mathbf{E}_0 \cos\left[\left(\sqrt{\frac{\mu\sigma\omega}{2}}z - \omega t\right)\right] \exp\left[-\sqrt{\frac{\mu\sigma\omega}{2}}z\right]$$

(c) We thus see that the amplitude of the electric field decreases exponentially as z increases. The depth δ, called the skin depth, at which the field decays to $1/e$ of its value at $z = 0$ is therefore

$$\delta = \sqrt{\frac{2}{\mu\sigma\omega}}.$$

Thus, in a good conductor, high frequency electromagnetic currents are restricted to a thin layer (the "skin") at the surface of the conductor. For a typical metal, taking $\sigma \sim 10^7(\Omega m)^{-1}, \mu \sim 10^{-6}N/A^2$, we get $\delta = 10^{-8}m$ for $\omega = 10^{15}$ s^{-1} (optical frequencies).

8. Problem-8

Question: An infinitely long straight non-magnetic conductor (wire) with a circular cross-section of radius a carries a steady current I. The current is distributed uniformly over the cross-section of the wire. The conductivity of the wire is σ. Find the rate at which energy flows into unit length of the conductor.

Solution:

Inside the conductor, the magnetic field is

$$\mathbf{B} = \frac{\mu_0 I \varrho}{2a^2}\hat{\varphi}$$

The electric field is obtained using $\mathbf{J} = \sigma\mathbf{E}$. We obtain

$$\mathbf{E} = \frac{I}{\pi a^2 \sigma}\hat{\mathbf{z}}$$

The Poynting vector $\mathbf{S}$ at the surface of the conductor, i.e., $\varrho = a$, is given by

$$\mathbf{S} = \mathbf{E} \times \frac{\mathbf{B}}{\mu_0} = -\frac{I^2}{2\pi^2 a^3 \sigma}\hat{\varrho}$$

The energy flux out of the surface of the conductor of length L is (with da $= dzad\varphi\hat{\varrho}$)

$$\int \mathbf{S} \cdot \mathbf{da} = -\frac{I^2}{2\pi^2 a^3 \sigma} \times 2\pi a L = -\frac{I^2 L}{\pi a^2 \sigma} = -I^2 R$$

where $R = \frac{L}{\pi a^2 \sigma}$ is the resistance of the conductor. Therefore, the energy flow into unit length of the conductor is $+\frac{I^2}{\pi a^2 \sigma}$.

9. Problem-9

Question: A beam of protons has a circular cross-section. Each proton has a velocity $\mathbf{v}$, and the beam constitutes a current I. Find the direction and magnitude of the Poynting vector $\mathbf{S}$ outside the beam, at a distance r from the axis of the beam.

Solution:

We need to compute the electric and magnetic fields due to the beam of protons. Choose

the beam to be aligned with the z-axis.

Then, we have that $\mathbf{H} = \frac{I}{2\pi\varrho}\hat{\varphi}$ (the magnetic field due to infinitely long wire carrying current I) and $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0\varrho}\hat{\varrho}$, with $\lambda = I/v$ (the electric field due to an infinite line charge λ).

The Poynting vector is

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{I^2}{4\pi^2\epsilon_0 v\varrho^2}\hat{\mathbf{z}} = \frac{I^2}{4\pi\epsilon_0\left(\pi v^2\varrho^2\right)}\mathbf{v}$$

Thus, we see that $\mathbf{S}$ is parallel to $\mathbf{v}$ consistent with the (other) interpretation of $(1/c^2)$ times the Poynting vector as a momentum density.
Check that the pre-factor has the dimension of momentum per unit volume.
[1]

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